Proficiency Modulates L1-Interference in L2 Prediction: Evidence from Chinese-Japanese Bilinguals in Classifier Prediction

People often predict upcoming language during comprehension both in L1 and L2, and bilingual prediction was facilitated when the word was a cognate versus a non-cognate (e.g., Ito et al., 2025). Crucially, this co-activation is proficiency-dependent, showing reduced facilitation effect in highly proficient L2 (Van Hell & Tanner, 2012). It can be attributed to strengthened direct L2 conceptual links rather than reliance on L1 (RHM), or stronger inhibitory control over L1 activation to resolve the interference (BIA-d) in highly proficient bilinguals (Grainger et al., 2010). We tested this inhibitory account using the Chinese-Japanese classifier system. Classifiers encode semantic properties of upcoming nouns (Adams & Faires, 1973) and allow for manipulation over form–meaning mappings, as there are Cognates (sharing similarity in both orthographic form and meaning in two languages), Non-cognates, or False Cognates (words sharing orthographic forms but differing in semantic representation) between Chinese and Japanese. We investigated whether L1-based semantic distraction overrides L2 predictive cues, and specifically, whether high proficiency enables the suppression of this L1 interference as predicted by BIA-d.

Two visual-world eye-tracking experiments tested a preliminary sample of 26 Chinese-Japanese bilinguals. Participants listened to simple Japanese sentences (structure: *Locative – Numeral - Classifier - GENITIVE - Noun - Existential Verb*) (e.g., *Mise no tana no ue ni wa - ichi - mai - no - shatsu - ga aru*. 'On the shelf in the store, one <classifier-flat object> shirt exists.') accompanied by visual scenes (Figure 4). Classifiers were manipulated: (1) Cognate (same form/function, e.g., *dai*, *teki*, *tsubu*), (2) Non-cognate (L2 specific, e.g., *sara*, *ken*, *wa* in L2 only), (3) False Cognate (same form but distinct semantic restriction, e.g., *mai*, *hon*, *piki*), and (4) General (non-predictive baseline *ko*). In Experiment 1, each display contained four images, one target and three unrelated pictures. Experiment 2 used the same materials and procedure, and introduced a critical manipulation: in the false-cognate condition, one unrelated picture was replaced with an L1 competitor corresponding to the noun selected by the false-cognate classifier in Chinese but not Japanese.

In Experiment 1 (N = 12), a significant main effect of classifier condition was observed, $F_{(3, 35.63)} = 3.13$, $p < .05$, conditional $R^2 = .131$. Prediction was robust for Non-cognate (divergence at -119ms relative to noun onset) but absent in the False-cognate condition (233ms), suggesting L1 interference disrupts L2 cue utilization (Figure 1). Experiment 2 (N = 14), replicated the significant main effect of condition (see Figure 2), $F_{(3, 35.43)} = 10.28$, $p < .001$, conditional $R^2 = .133$. The Non-cognate condition elicited the strongest prediction effect (divergence at -143ms), while the False-cognate condition showed no prediction (163ms). Further, the false cognate condition revealed a Proficiency × Picture Type interaction, $\chi^2(1) = 12.35$, $p < .001$ (Figure 3). High-proficiency speakers looked significantly more to the target than the L1 competitor, $z = -4.50$, $p < .0001$, while low-proficiency speakers showed no such difference, $z = -0.15$, $p = 0.88$.

Overall, these preliminary results show that classifier-based prediction depends on cognate status in L2 processing, modulated by proficiency. Consistent with BIA-d, false cognates disrupted anticipatory looks, indicating that L1 semantic activation may overshadow L2 classifier cues, particularly for low-proficiency speakers, who were unable to suppress L1 semantic activation. In contrast, high-proficiency speakers successfully inhibited this interference, achieving stable L2 prediction. Unexpectedly, we found no cognate advantage for both groups. While this warrants caution given the preliminary sample size, it suggests that in systems requiring complex conceptual categorization more than single lexical item activation, which is common in classifier systems, form overlap may offer little predictive benefit for anticipatory processing.

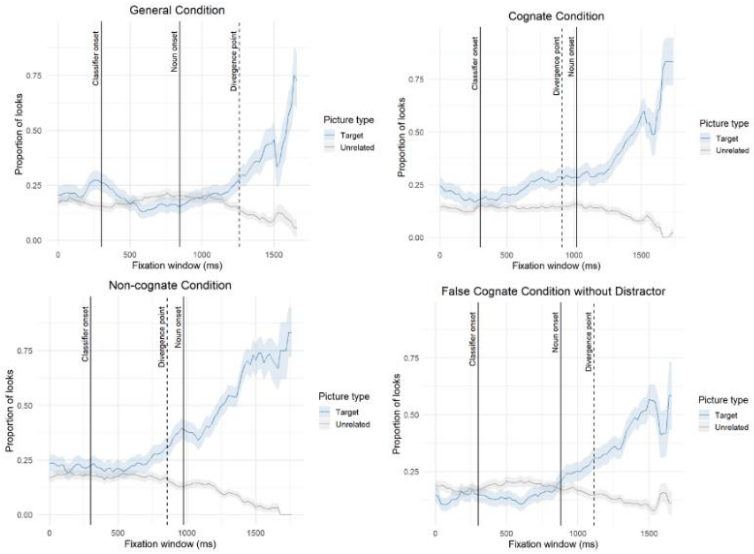

Figure 1. Time course of looks to the target versus non-targets across the four classifier conditions in Experiment 1.
Note. **The divergence point is the earliest time when target and non-target fixations differ. Negative values indicate early target looks before noun onset; positive values indicate late target looks after noun onset.**

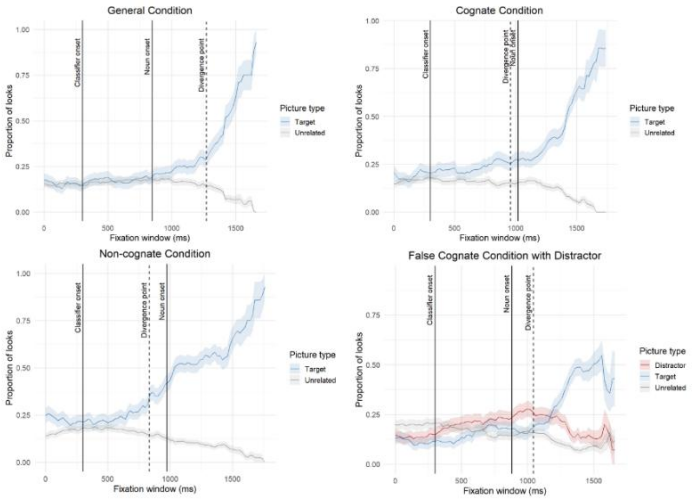

Figure 2. Time course of looks to the target versus non-targets across the four classifier conditions in Experiment 2.

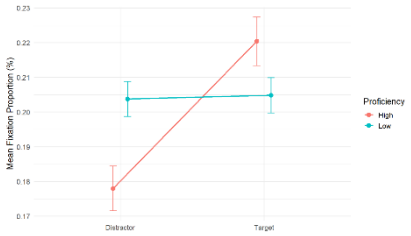

Figure 3. Mean Fixation Proportions to Targets and Distractors in the False-Cognate Condition by Proficiency Group.

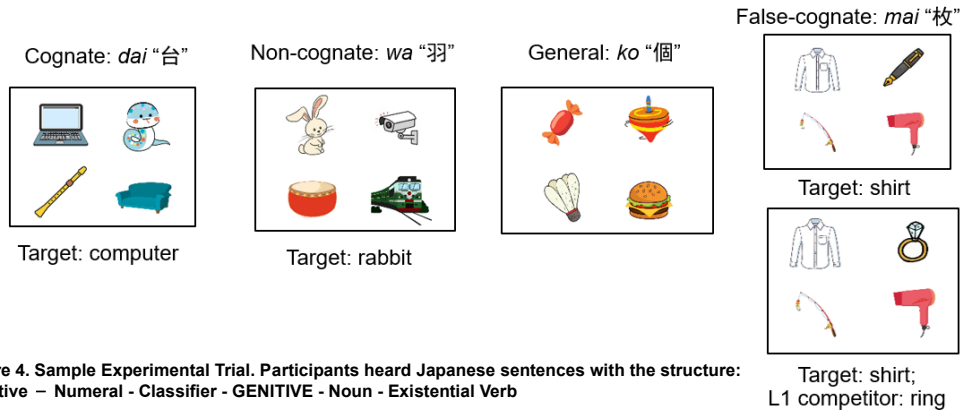

Figure 4. Sample Experimental Trial. Participants heard Japanese sentences with the structure: Locative – Numeral - Classifier - GENITIVE - Noun - Existential Verb

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